

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

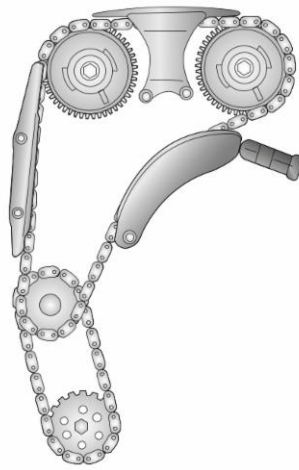
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

General

Timing chain layout



Removal

Remove the cowl lower grille

Disconnect the battery earth cable

Remove the engine together with the gearbox

If applicable: Remove the transfer box.

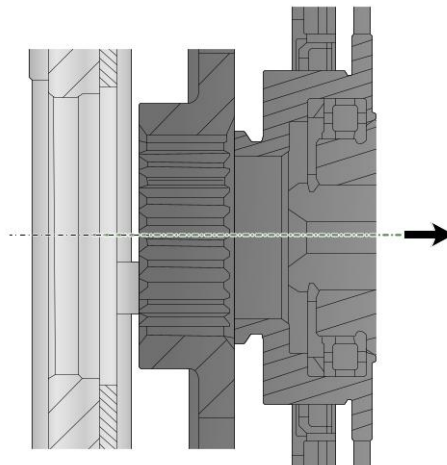
Vehicles with dual-clutch transmission:

Disconnect the gearbox cooling lines

Remove the engine/gearbox attaching bolts

Check that the position is correct as shown

Remove the dual-clutch transmission



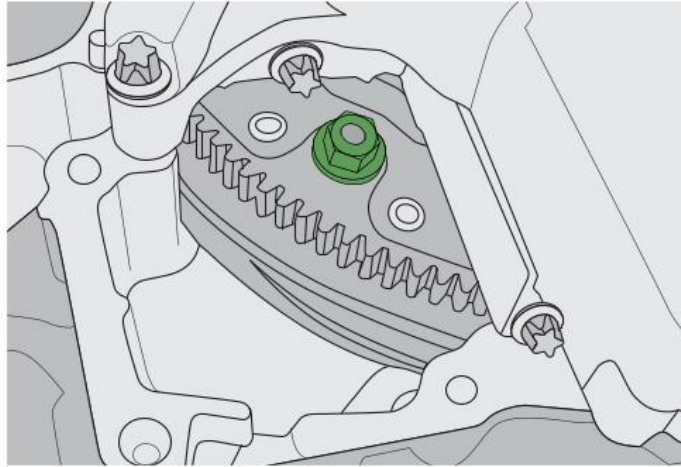
Vehicles with automatic transmission:

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Fit the crankshaft turning tool (11 6 480)

Note: Turn the engine clockwise

Remove the torque converter attaching nuts ((x6))



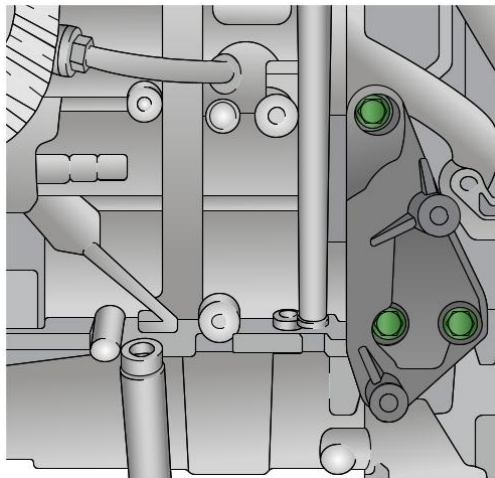
Separate the gearbox from the engine

All vehicles:

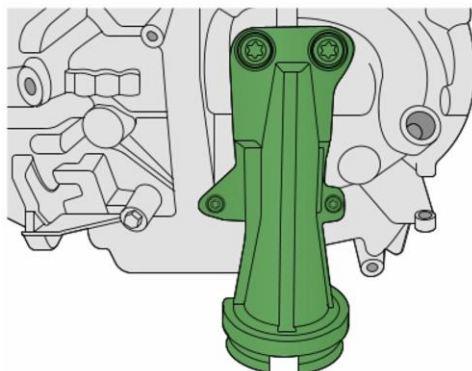
Fit a lifting eye to the cylinder head ((21.5 Nm)) (2 459 012)

Note: Use a hoist to lift the engine (11 0 070)

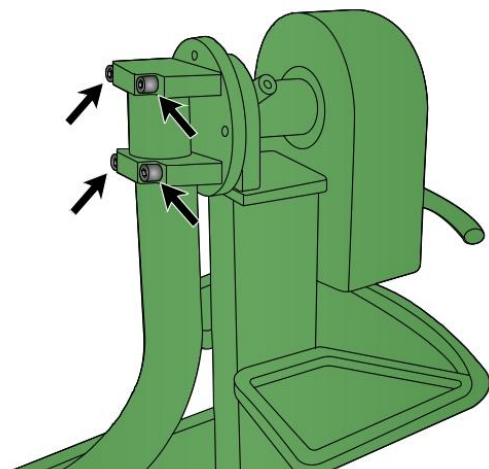
Remove the engine support bracket



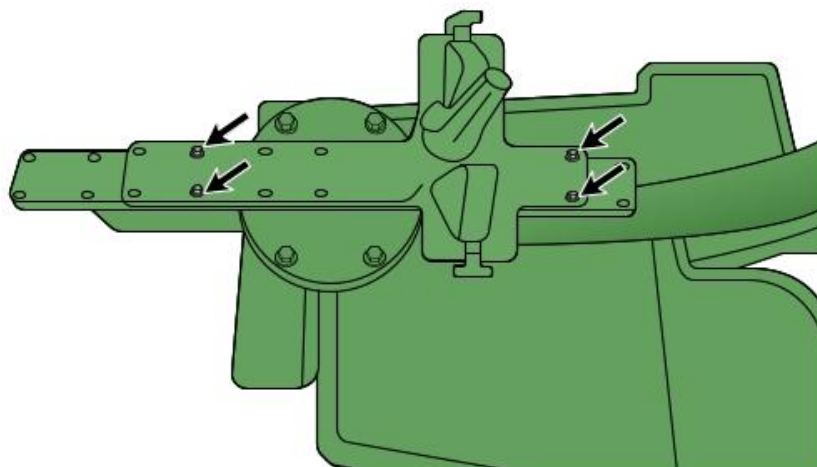
Fit the special tool (2 455 010)

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Assemble the special tool as shown (00 2 320, 00 2 300)



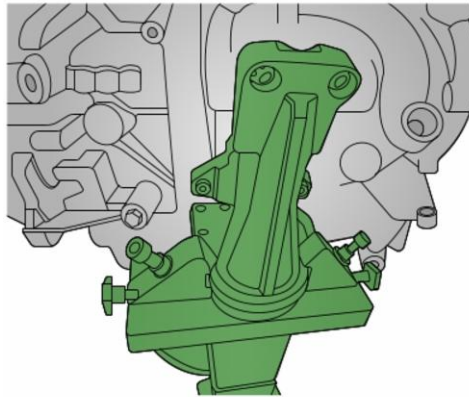
(2 413 082)



Lower the engine

Tighten the special tool (2 413 082, 2 455 010)

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Remove the cylinder head

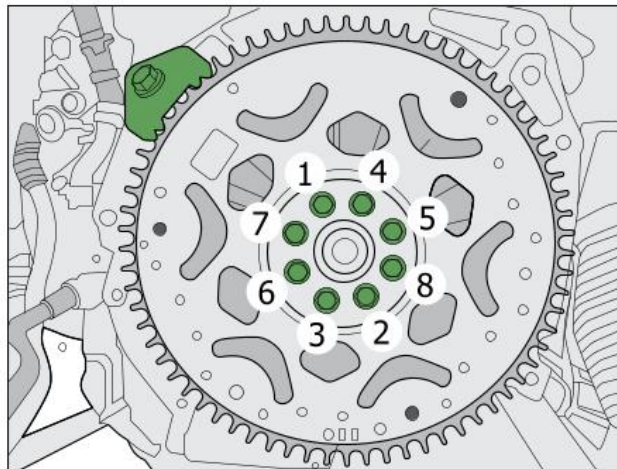
Vehicles with dual-clutch transmission:

Fit the locking tool. With washer: (5 mm).

Remove the drive plate (11 4 180)

Vehicles with automatic transmission:

Fit the locking tool. With washer: (5 mm).



Remove the flexplate

All vehicles:

Disconnect the oil level sensor

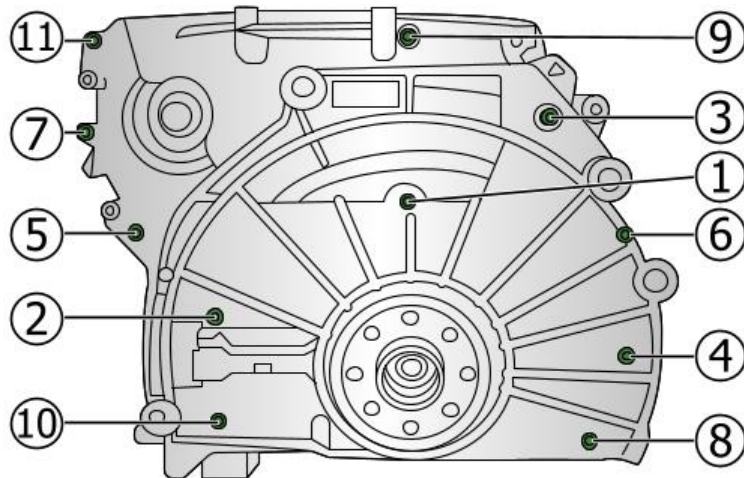
Remove the lower torque reaction rod

Remove the oil sump

Disconnect the crankshaft position sensor

Remove the bolts in the reverse order to that shown ((11 -> 1))

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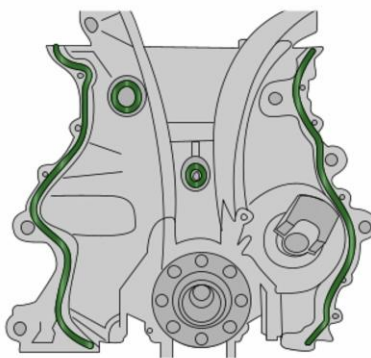
- Remove the timing chain cover
- Remove the chain tensioner guide
- Remove the chain guide
- Remove the timing chain
- Remove the oil pump gearwheel
- Remove the oil pump chain

Installation

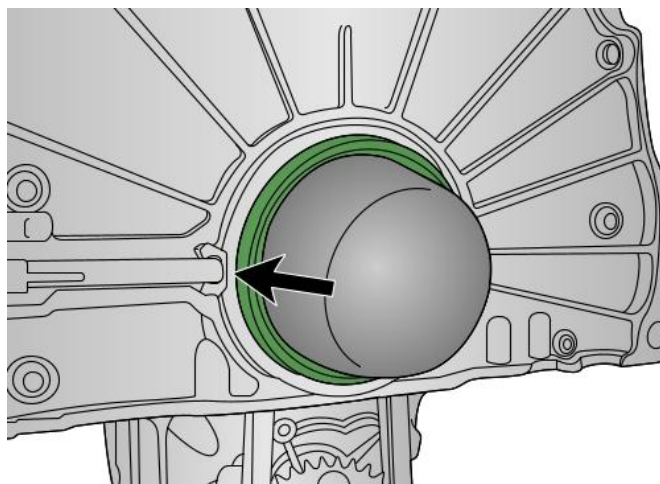
- Note: Clean off the sealing surfaces
- Installation is in the reverse order of removal

Timing chain cover:

- Remove and discard the crankshaft oil seal
- Check the dowel pins for damage
- Apply sealant as indicated ((2.0 - 2.5 mm))

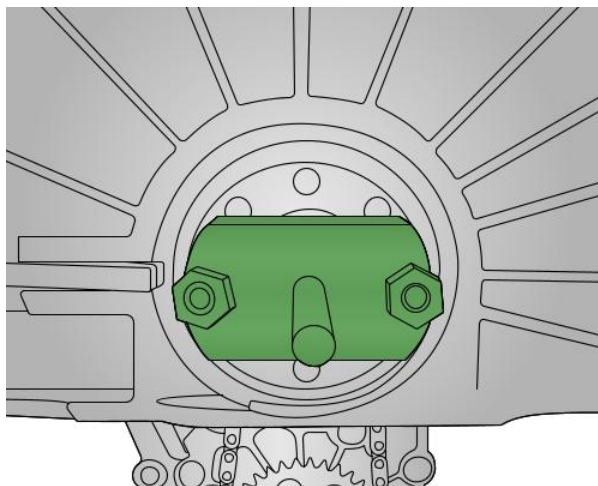


- Note: The cover must be fitted within 10 minutes of applying the sealant
- Refit the timing cover
- Note: Use new bolts
- Reconnect the crankshaft position sensor
- Fit the special tool (11 8 815)
- Note: Do not apply lubricant
- Fit the crankshaft seal over the special tool
- Note: Do not touch the lip of the seal

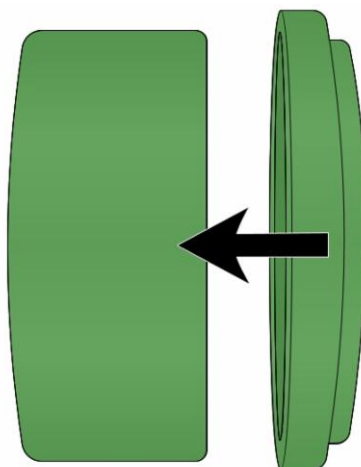
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Remove the special tool (11 8 815)

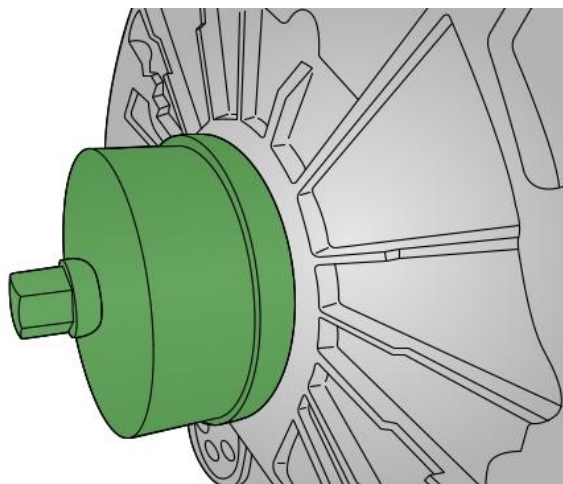
Fit the special tools (11 9 181, 11 9 182, 11 9 183, 11 9 184, 11 9 185)



Assemble the special tool as shown (2 360 215)



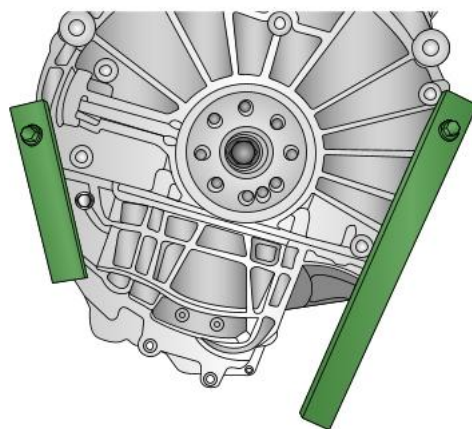
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Tighten the special tool bolt until full surface contact is made

Refit the oil sump

Use the special tools (11 8 780)



Further refitting is in the reverse order of removal

Reconnect the battery earth cable

Torque settings

Oil pump gearwheel

Oil pump gearwheel (Renew the bolt (left-hand thread))

Stage 1 5 (Nm)

Stage 2 90 (°)

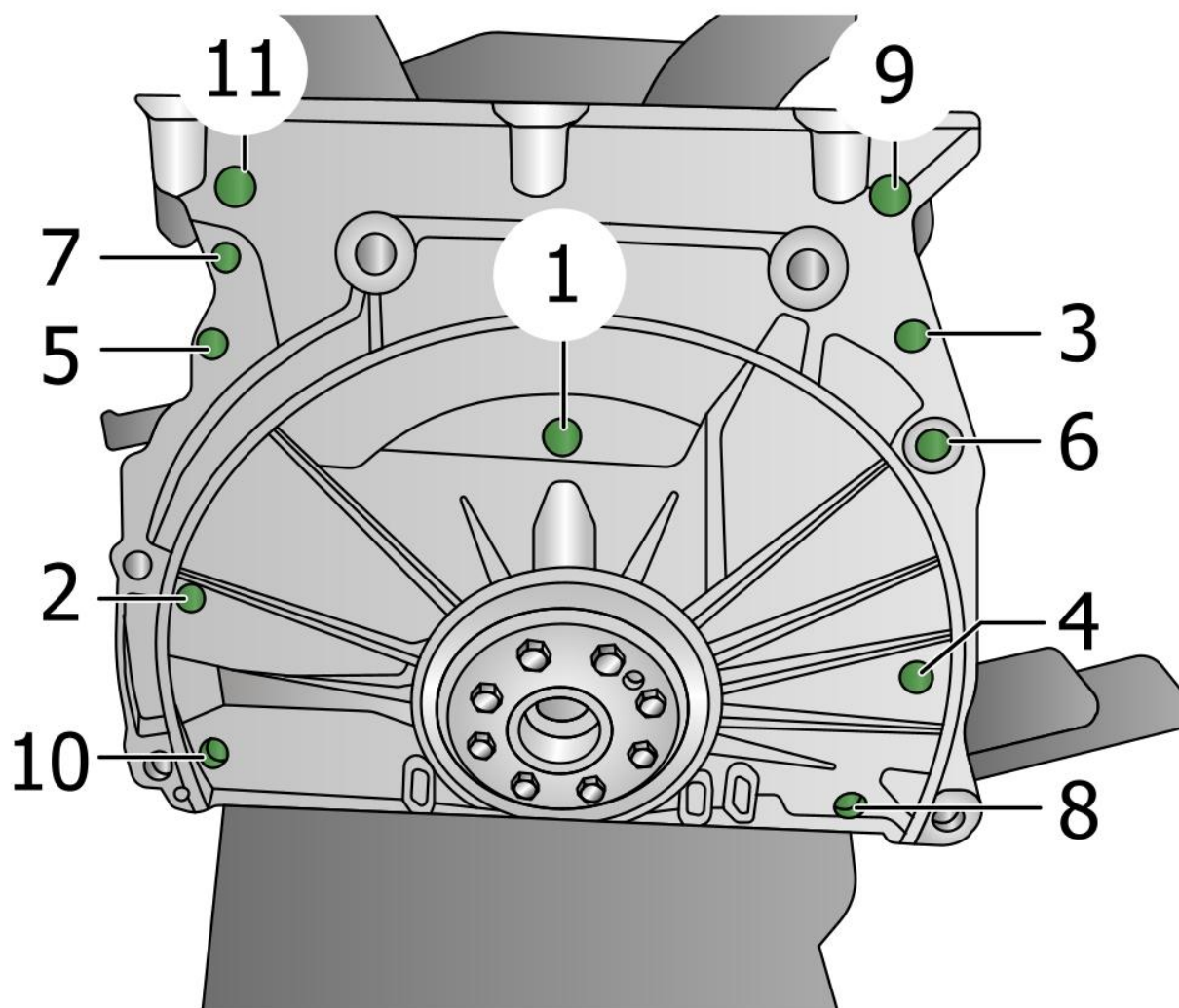
Timing chain tensioner guide

Timing chain tensioner guide (Renew the bolt) 22 (Nm)

Timing cover

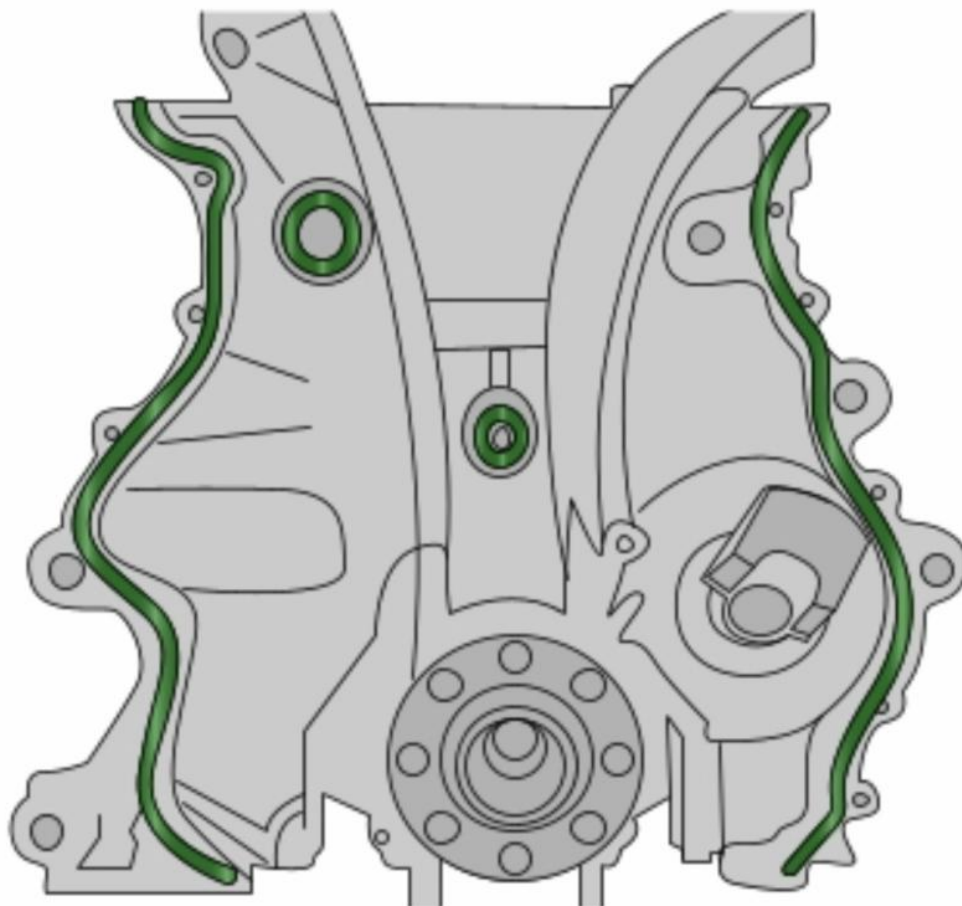
Timing cover (Renew the bolts)

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Clean the contact surfaces

Apply sealant (BMW 83 19 0 404 517; Loctite 5970) 2.0 - 2.5 (mm)

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The component must be fitted within 10 minutes of applying the sealant

Wait for 25 minutes before starting the engine

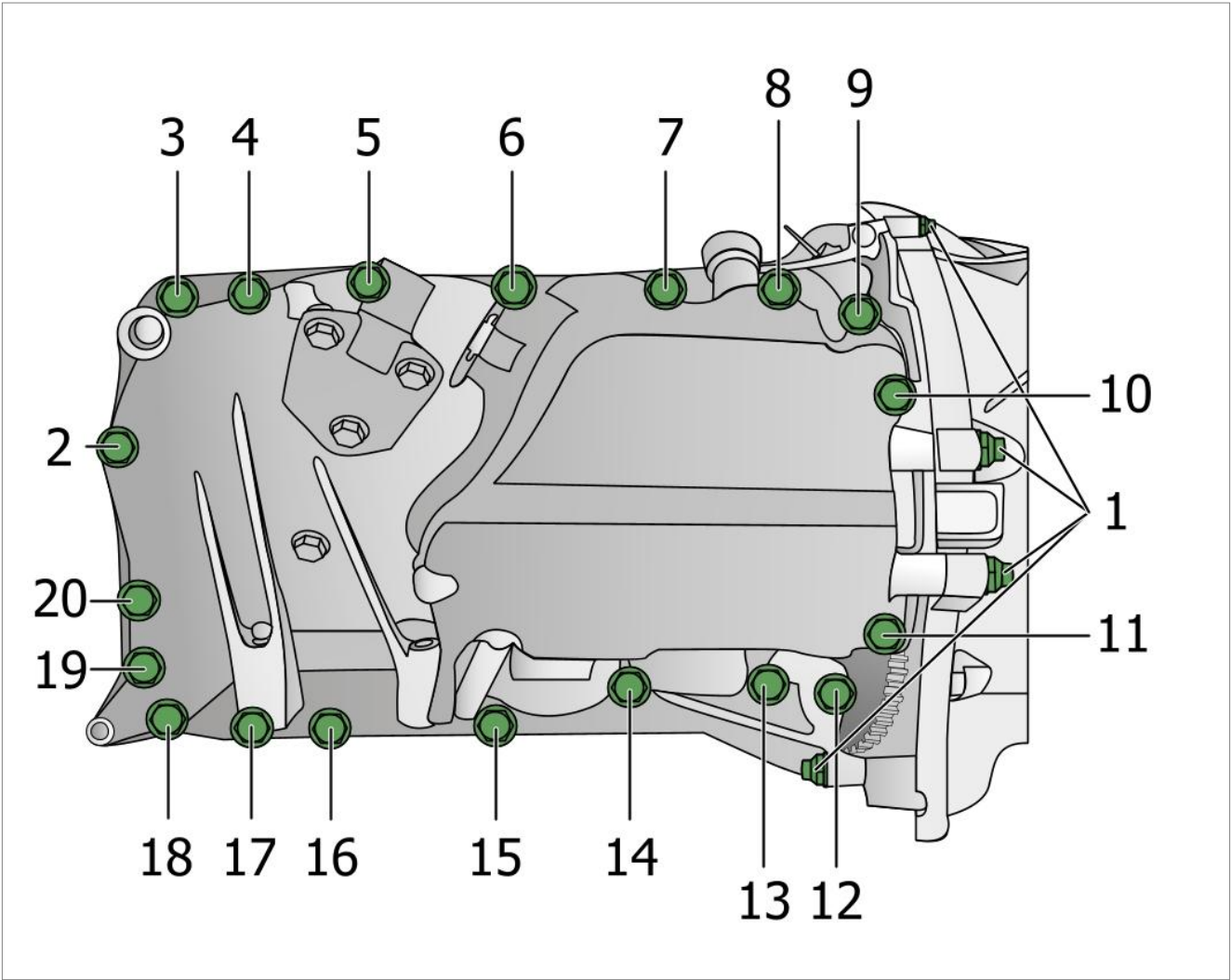
Stage 1 8 (Nm)

Stage 2 90 (°)

Oil sump

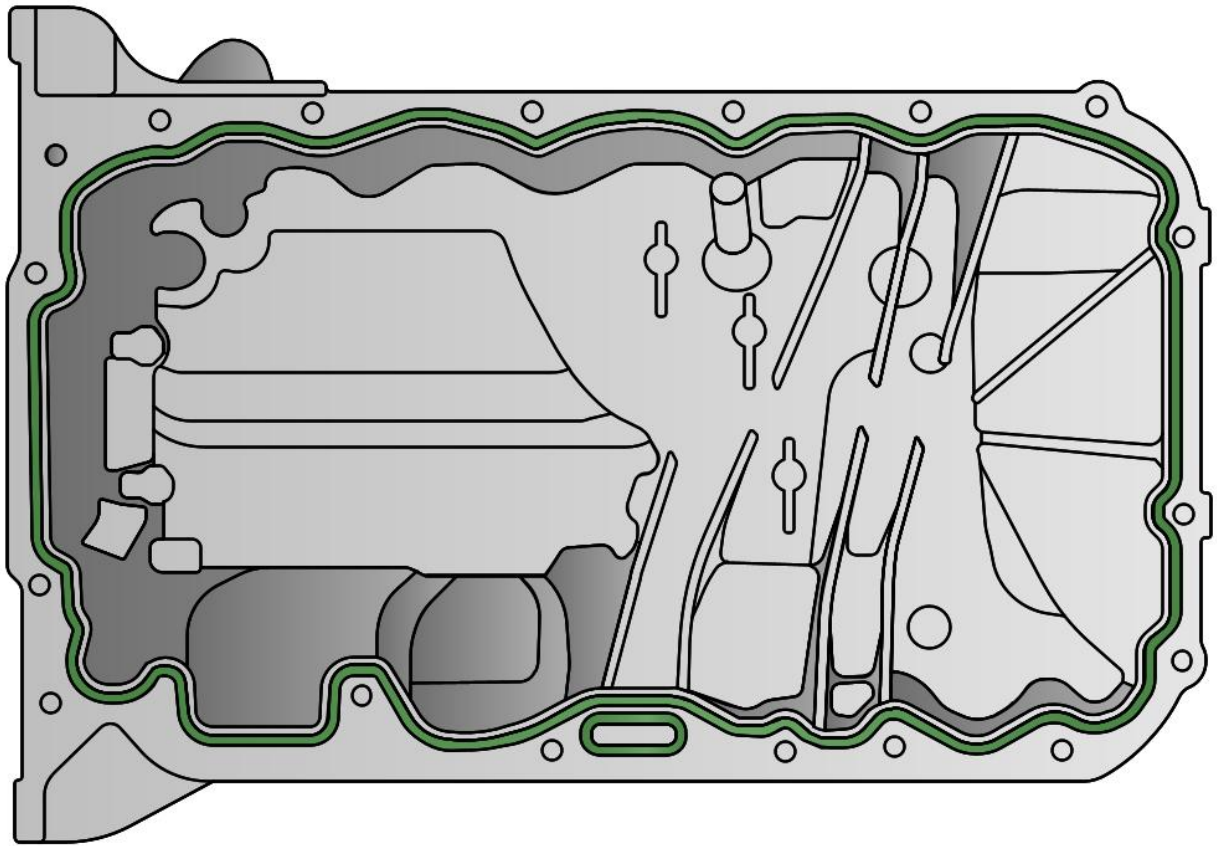
Sump (Clean the contact surfaces)

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Note: Bolts have different lengths

Apply sealant (BMW 83 19 0 404 517; Loctite 5970) 2.0 - 2.5 (mm)

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The component must be fitted within 10 minutes of applying the sealant

Wait for 25 minutes before starting the engine

Stage 1 (Tighten the bolts finger-tight)

Stage 2 (Bolts 2 - 20) 24 (Nm)

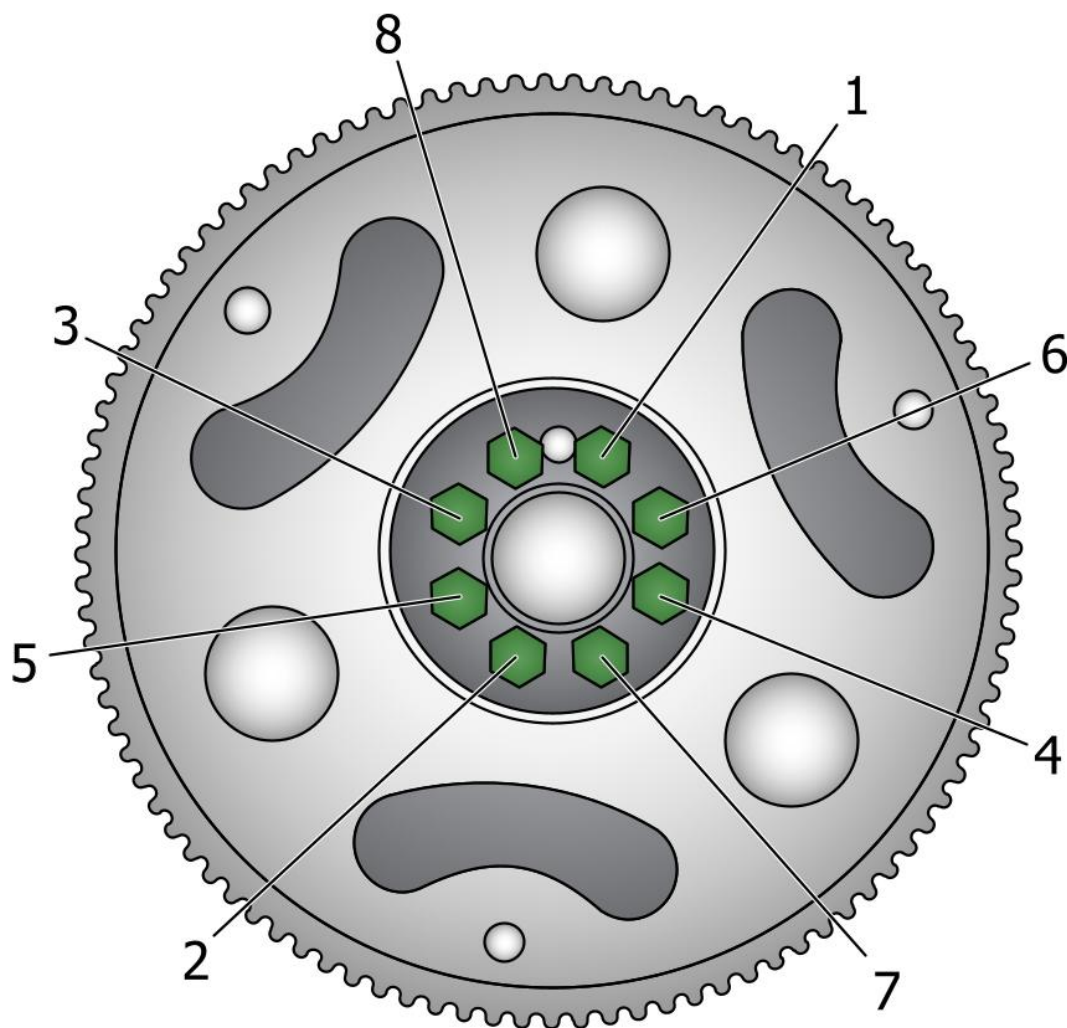
Stage 3 (Bolts 1) 19 (Nm)

Drive plate

Flexplate

Flexplate (Renew the bolts)

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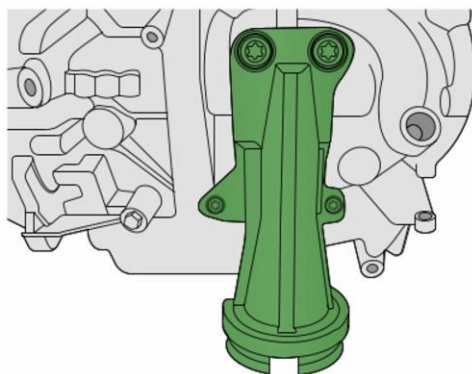
Clean the contact surfaces

Stage 1 (Tighten the bolts finger-tight)

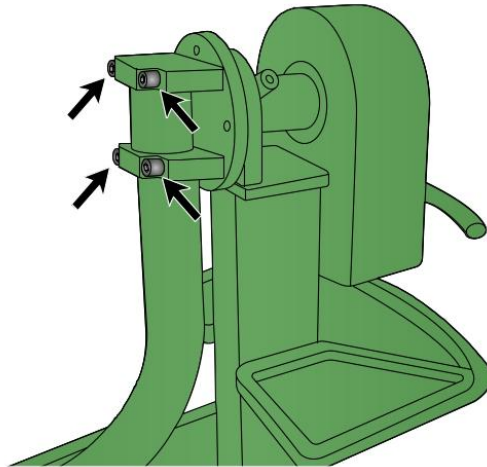
Stage 2 45 (Nm)

Stage 3 45 (°)

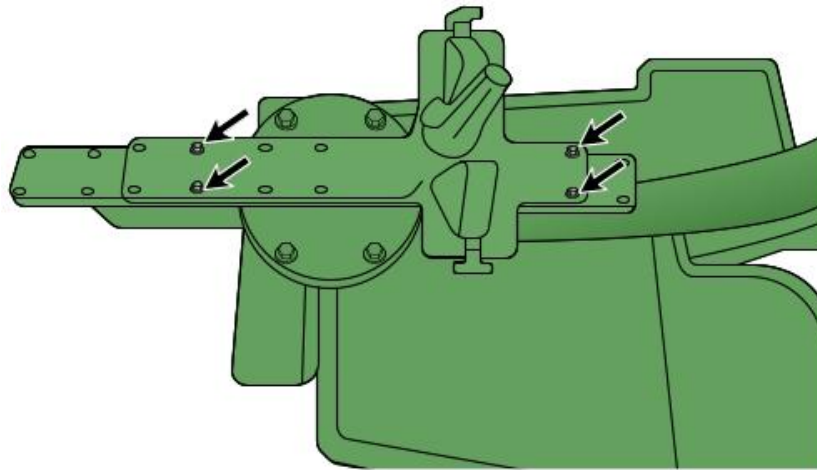
Special tool: (38 Nm) (2 455 010)



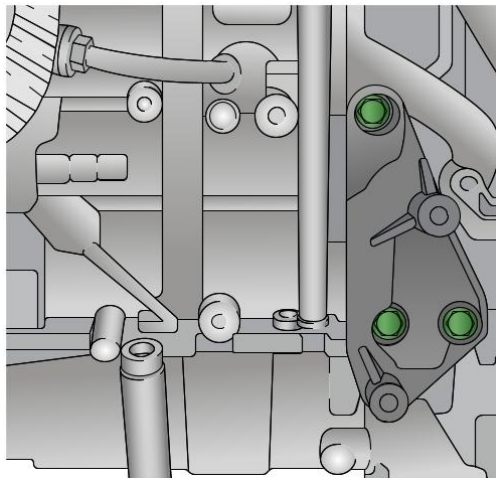
Special tool: (130 Nm) (00 2 320)

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Special tool: (160 Nm) (2 413 082)



Engine support bracket: (38 Nm)



Engine/gearbox attaching bolts:

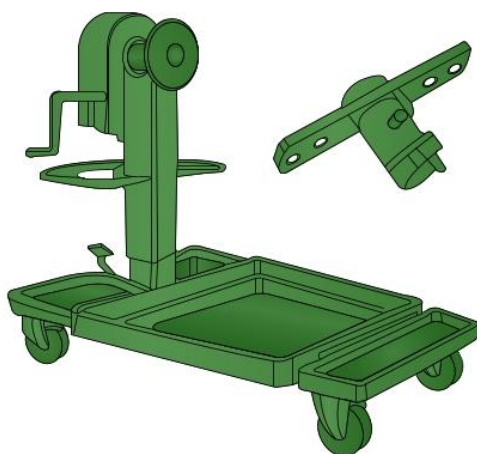
M8: (19 Nm)

M12: (66 Nm)

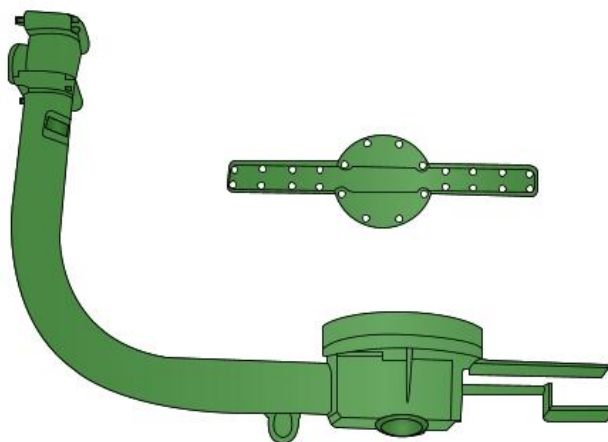
Special tools

Engine support: (00 2 300)

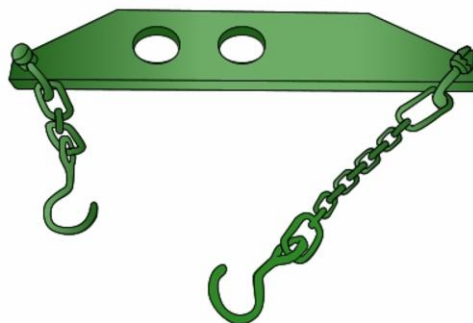
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Engine support: (00 2 320)

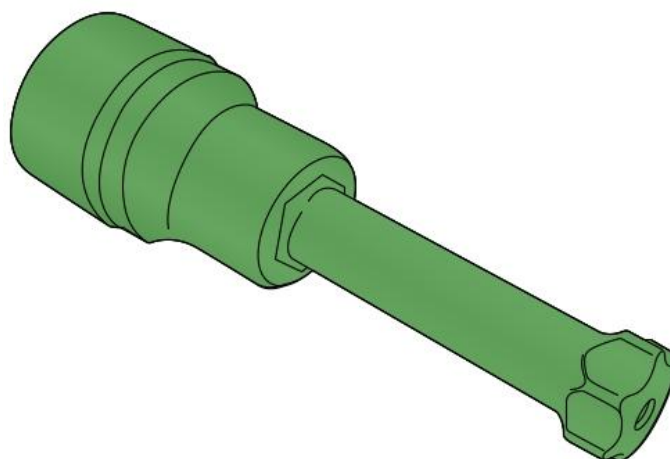


Lifting tool: (11 0 070)



Socket: (11 4 180)

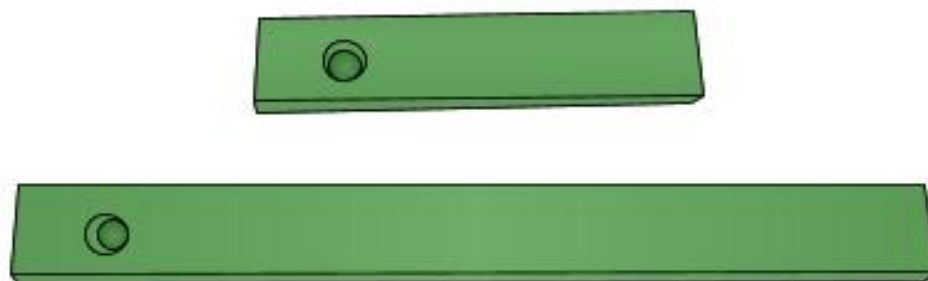
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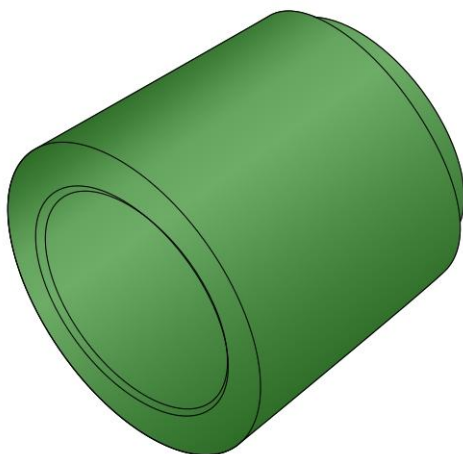
Socket: (11 6 480)



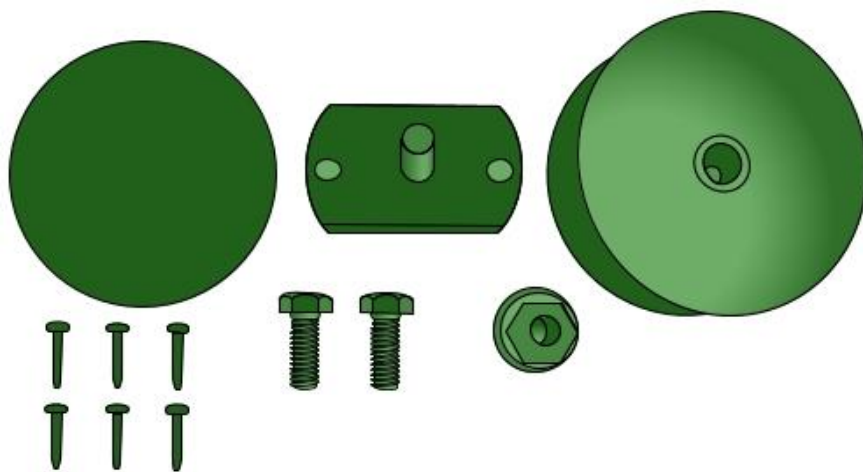
Special tools: (11 8 780)



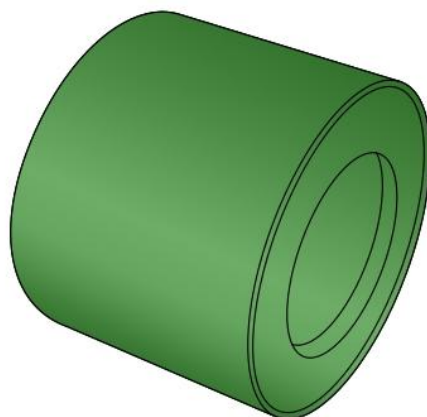
Oil seal fitting tool: (11 8 815)

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Oil seal fitting tool: (11 9 181, 11 9 182, 11 9 183, 11 9 184, 11 9 185)

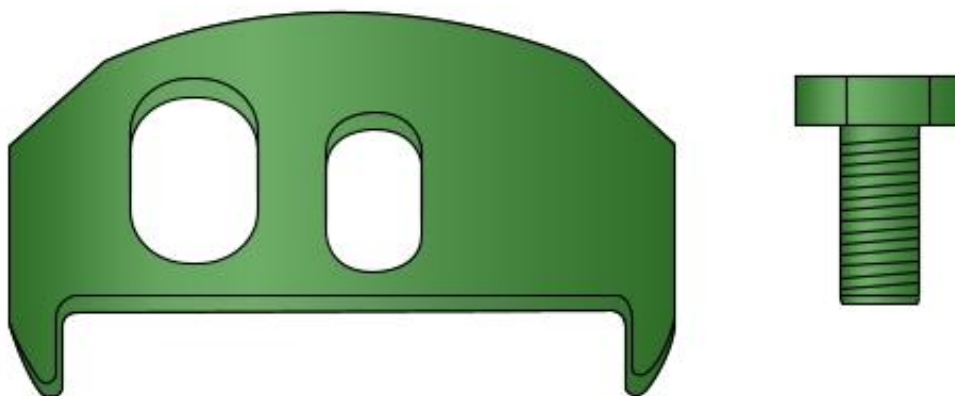


Bush: (11 9 265)

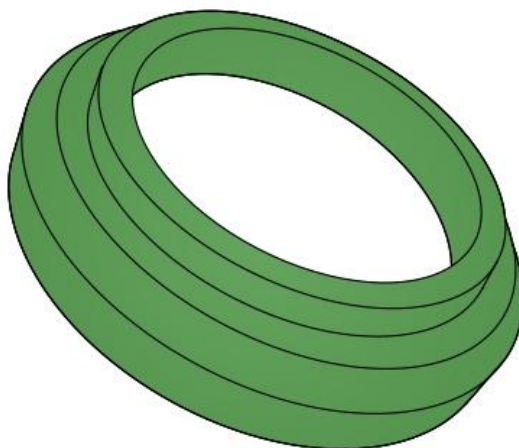


Flywheel locking tool: (11 9 811)

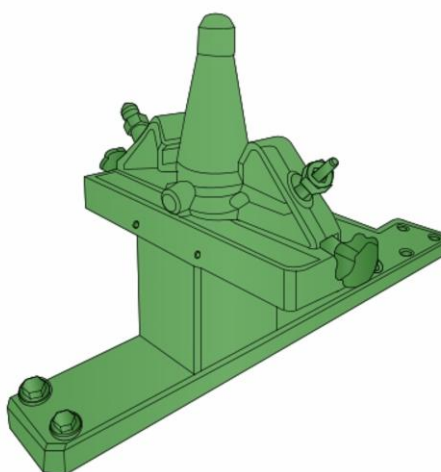
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Adaptor: (2 360 215)

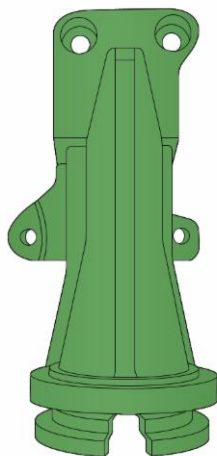


Engine support: (2 413 082)



Engine support tool: (2 455 010)

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Lifting eye: (2 459 012)

